

Guide 96 — Electrical and Electronic Engineering Technician

Version: 2.0 controlled working master

Language: English

Primary U.S. benchmark: O*NET-SOC 17-3023.00 — Electrical and Electronic Engineering Technologists and Technicians

Canada comparison: NOC 22310 — Electrical and electronics engineering technologists and technicians

Colombia comparison: CUOC 31141 — Técnicos en electrónica

What this career is

Electrical and electronic engineering technicians support the design, testing, production, operation, troubleshooting, maintenance and improvement of electrical and electronic equipment and systems.

The work sits between theory and hands-on technical practice. A technician may read schematics, assemble prototypes, prepare test setups, record measurements, inspect components, identify faults, document changes, support calibration, maintain equipment, assist engineers, and help keep production or laboratory systems working reliably.

The exact role varies widely. One employer may focus on electronics manufacturing. Another may focus on instrumentation, utilities, communications, industrial controls, laboratory testing, medical equipment, transportation systems, aerospace, renewable energy, or field service.

What this career is not

A technician title does not automatically make someone:

- a licensed professional engineer;
- an electrician authorized to perform regulated installation work;
- a qualified person for energized electrical work;
- an inspector or certifying authority;
- a calibration laboratory signatory;
- a cybersecurity administrator;
- an equipment manufacturer's authorized representative.

Those authorities depend on law, employer policy, training, certification, licensing, supervision, equipment, jurisdiction and assigned duties.

The safe rule is simple: perform only work for which you are trained, authorized and properly equipped.

Why employers use technicians

Engineers and technical organizations need reliable people who can connect design intent to physical systems.

Technicians help by:

- building and modifying prototypes;
- reading schematics and wiring diagrams;
- installing or replacing approved components within assigned scope;
- setting up test fixtures;
- recording measurements;
- identifying patterns in failures;
- supporting preventive and corrective maintenance;
- documenting configurations and revisions;
- checking workmanship and quality;
- supporting production and field teams;
- preparing technical reports;
- escalating conditions outside their authority.

Strong technicians are valuable because they do not merely “fix things.” They work methodically, protect people and equipment, preserve traceability, and know when to stop and escalate.

Common work settings

Possible settings include:

- electronics manufacturing;
- engineering laboratories;
- utilities;
- industrial plants;
- communications companies;
- transportation systems;
- aerospace and defense suppliers;
- renewable-energy facilities;
- instrumentation and controls;
- medical-device or biomedical support environments;
- consulting firms;
- government facilities;
- field-service organizations;
- research and development teams.

Some jobs are mostly bench/lab work. Others include production floors, equipment rooms, customer sites or industrial facilities.

Typical entry preparation

In the United States, BLS says this occupation typically requires an associate degree. O*NET places it in Job Zone Three, where vocational training, related experience or an associate degree is common.

Common program names include:

- electrical engineering technology;
- electronics engineering technology;
- electronics technology;
- instrumentation technology;
- industrial electronics;
- automation;
- mechatronics;
- industrial controls;
- calibration/metrology;
- telecommunications technology.

Employers may also hire through military technical training, apprenticeships, employer development programs, or related industrial experience.

Core technical foundation

A useful foundation includes:

- basic algebra and unit conversion;
- scientific notation;
- voltage, current, resistance and power concepts;
- series and parallel relationships;
- AC and DC concepts;
- analog and digital signals;
- frequency and timing;
- component identification;
- schematic symbols;
- grounding and shielding concepts;
- sensors and transducers;
- relays and contactors;
- power supplies;
- basic semiconductor concepts;
- connectors and cabling;
- measurement uncertainty awareness;
- documentation and change control.

The goal is not memorizing formulas. The goal is understanding what a measurement or diagram means and when a result does not make sense.

Reading schematics and technical drawings

Technicians often work from:

- electrical schematics;
- wiring diagrams;
- block diagrams;
- ladder diagrams;
- P&IDs in industrial settings;
- cable schedules;
- connector pinouts;
- assembly drawings;
- bills of materials;
- equipment manuals;
- manufacturer data sheets;
- engineering change notices.

A disciplined reading process asks:

1. What revision is this document?
2. What equipment or configuration does it apply to?
3. What are the reference designators and signal names?
4. What assumptions or prerequisites are stated?
5. Is the drawing approved for the work being performed?
6. Does the physical equipment match the documentation?

Never silently “correct” an approved drawing because the field equipment looks different. Record the discrepancy and use the employer’s change-control process.

Test equipment and measurement thinking

Depending on the role, equipment may include:

- digital multimeters;
- oscilloscopes;
- signal generators;
- power supplies;
- logic analyzers;
- current probes;
- insulation or continuity test equipment;
- frequency counters;

- calibration standards;
- temperature/pressure/process instruments;
- computer-based diagnostic tools.

A technician should understand that every measurement has limits. Instrument rating, range, accuracy, resolution, calibration status, probe condition, connection method and environment can affect the result.

Safety boundary: this guide does not instruct readers to test energized electrical circuits. OSHA requires qualified-person controls for electrical testing work on circuits or equipment. Learning should default to simulations, training boards, deenergized equipment, low-energy electronics specifically designed for instruction, or supervised laboratories with established procedures.

Troubleshooting as a controlled process

Good troubleshooting is evidence-based.

A general process is:

1. define the reported symptom;
2. confirm the correct equipment/configuration/revision;
3. review safety and authorization boundaries;
4. inspect for obvious approved-to-check conditions;
5. compare observations with known-good documentation or expected behavior;
6. isolate the problem logically rather than replacing parts at random;
7. record what was tested and what changed;
8. verify the repair or correction under the approved procedure;
9. return documentation/configuration to the required state;
10. escalate unresolved or unsafe conditions.

Do not bypass safety devices or interlocks simply to make a symptom disappear.

Components and assemblies

Common learning topics may include:

- resistors;
- capacitors;
- inductors;
- diodes;
- transistors;
- integrated circuits;
- optocouplers;

- connectors;
- switches;
- relays;
- fuses and protective devices;
- sensors;
- printed circuit boards;
- motors and actuators;
- power supplies;
- communications interfaces.

You do not need to become a component-design engineer to be employable. Entry-level value often comes from correctly identifying parts, following documentation, measuring safely, documenting results, recognizing abnormal conditions and escalating appropriately.

Analog, digital and embedded systems

Modern equipment often combines analog, digital and embedded functions.

Useful concepts include:

- signal level and range;
- noise;
- grounding/shielding;
- sampling;
- logic states;
- timing;
- analog-to-digital and digital-to-analog conversion;
- microcontrollers;
- firmware versions;
- communications protocols;
- interfaces between sensors, controllers and actuators.

A technician should avoid assuming that a hardware symptom is always a hardware failure. Configuration, firmware, software, communications, power quality and environmental conditions can also matter.

Instrumentation and industrial controls

Some jobs overlap with instrumentation and controls.

Possible topics include:

- process sensors;
- transmitters;
- control loops;

- PLC concepts;
- I/O modules;
- alarms;
- interlocks;
- variable-frequency drives;
- human-machine interfaces;
- industrial networks;
- calibration records;
- preventive maintenance;
- configuration backups.

These systems can affect production and safety. Do not change a parameter, bypass an interlock, force an output or connect a programming device to production equipment unless the procedure, authorization and safety conditions permit it.

Calibration and traceability

Calibration is more than “adjusting until the number looks right.”

A controlled calibration environment may require:

- approved procedures;
- traceable standards;
- environmental conditions;
- calibration intervals;
- as-found and as-left data;
- uncertainty awareness;
- equipment identification;
- technician authorization;
- review and sign-off.

A learner can practice the concepts with training instruments and simulated records, but should not claim formal calibration authority without the organization’s required system and credentials.

Manufacturing and quality support

Electronics-manufacturing technicians may support:

- incoming inspection;
- assembly;
- test stations;
- failure analysis;
- rework under approved procedures;
- quality checks;

- production troubleshooting;
- configuration control;
- engineering changes;
- equipment maintenance;
- documentation of defects and corrective actions.

Useful quality habits include:

- verify the correct work instruction;
- confirm revision and part number;
- record serial/lot identifiers when required;
- use calibrated tools when the procedure requires them;
- keep inspection and test results traceable;
- separate suspected nonconforming material;
- never falsify a passing result.

Soldering, rework and ESD awareness

Some electronics roles include soldering or circuit-board rework.

Training should cover:

- employer-approved soldering/rework procedures;
- ventilation/fume controls;
- hot-tool and burn hazards;
- eye protection where required;
- lead or chemical controls where applicable;
- electrostatic-discharge awareness;
- grounded work surfaces and handling methods when required;
- inspection criteria;
- authorization to repair specific equipment.

Do not practice on safety-critical, customer-owned, medical, aerospace or production hardware without explicit authorization.

Electrical safety — deenergization first

Electrical hazards can include shock, burns, arc events, fire, unexpected startup, stored energy and induced/backfeed voltage.

OSHA 29 CFR 1910.333 establishes that live parts to which an employee may be exposed generally must be deenergized before work on or near them unless the employer can justify an exception allowed by the standard.

Important boundaries:

- deenergized equipment that has not been properly locked/tagged must be treated as energized;
- stored electrical energy can remain after power is disconnected;
- capacitors and high-capacitance elements can retain hazardous energy;
- deenergization verification is qualified-person work under the standard;
- energized electrical work is restricted to qualified persons under the applicable requirements.

This guide therefore does not provide step-by-step energized-work instructions.

Lockout/tagout and stored energy awareness

Lockout/tagout is an employer-controlled hazardous-energy process, not a personal shortcut.

A learner should understand the purpose:

- identify energy sources;
- isolate equipment under the approved procedure;
- prevent unexpected reenergization or startup;
- address stored electrical and non-electrical energy;
- verify the required safe condition using authorized methods;
- control restoration to service.

Do not improvise lockout/tagout, remove another person's lock, or assume an open switch means equipment is safe.

Qualified-person testing boundary

OSHA 29 CFR 1910.334 states that only qualified persons may perform testing work on electric circuits or equipment.

The standard also requires test instruments, leads, cables, probes and connectors to be inspected and requires equipment to be appropriately rated for the circuit and environment.

For learners, safe alternatives include:

- circuit simulation;
- battery-powered educational kits designed for instruction;
- deenergized continuity exercises under approved conditions;
- instructor-supervised lab boards;
- virtual instrumentation;
- documentation-based fault analysis;

- manufacturer training simulators.

Never use this guide as permission to probe live mains, industrial panels or unknown-energy equipment.

Batteries and energy-storage awareness

Battery systems can present electrical, chemical, thermal and fire hazards.

Technicians should follow:

- manufacturer instructions;
- employer procedures;
- battery chemistry-specific requirements;
- approved charging/storage methods;
- short-circuit prevention;
- PPE and ventilation requirements;
- damaged-battery escalation procedures.

Do not puncture, heat, short, disassemble or experimentally charge unknown or damaged cells.

Professional engineering boundary

Technicians often contribute observations, measurements, prototypes and technical documentation used by engineers.

That contribution does not automatically authorize the technician to:

- approve engineering designs;
- sign/seal regulated engineering documents;
- determine code compliance as the authority having jurisdiction;
- accept safety-critical design changes;
- certify a system for service;
- override engineering controls.

When a decision affects regulated design, life safety, protection settings or engineering approval, follow the responsible professional and organizational authority.

Electrician and installation boundary

Engineering-technician work and electrician work can overlap, but they are not identical occupations.

Electrical installation, modification, inspection and energized-work authority can be regulated by jurisdiction, voltage, equipment type and workplace.

A technician should verify:

- whether an electrical license is required;
- whether the employer requires a particular trade qualification;
- whether the work is under a qualified electrician or engineer;
- whether site-specific authorization is required;
- whether energized work is prohibited or restricted.

Never assume a college diploma or electronics title automatically grants installation authority.

United States — occupation and pay

O*NET-SOC 17-3023.00

O*NET identifies the occupation as **Electrical and Electronic Engineering Technologists and Technicians**.

Current occupation-specific BLS 2025 wage values shown through O*NET are:

- 10th percentile: **\$49,510/year — \$23.80/hour**;
- 25th percentile: **\$61,610/year — \$29.62/hour**;
- median: **\$78,190/year — \$37.59/hour**;
- 75th percentile: **\$97,650/year — \$46.95/hour**;
- 90th percentile: **\$115,700/year — \$55.62/hour**.

These are national statistics, not salary promises. Pay varies by location, industry, clearance requirements, shift, experience, instrumentation/controls capability, equipment specialization and employer.

BLS Occupational Outlook Handbook context

BLS reports a **May 2024 median of \$77,180/year**, projects **1% employment growth from 2024 to 2034**, and estimates about **8,400 openings per year** on average over the decade.

BLS states that an associate degree is the typical entry-level education.

The 2024 OOH value and 2025 occupation-specific wage table use different reference periods. Do not average them.

United States — current non-government market context

Indeed reported approximately **\$30.67/hour** average base salary for the broader title **Electrical Technician**, based on about **9,000 salary observations from job postings over the prior 36 months**, updated August 10, 2026.

Indeed also reported approximately **\$27.58/hour** for **Electronics Technician**, based on about **8,400 salary observations**, updated August 9, 2026.

These are non-government market-title estimates. They are not equivalent to O*NET 17-3023.00 and are not guaranteed compensation.

United States — education and funded-training locators

CareerOneStop can help locate:

- electrical/electronics engineering technology programs;
- electronics technology programs;
- instrumentation/controls programs;
- automation and mechatronics programs;
- WIOA-eligible training providers;
- apprenticeships and other work-based learning.

WIOA funding is eligibility-based. Contact an American Job Center for current rules.

O*NET lists approved apprenticeship titles associated with the occupation, including Electrical Technician, Electronics Technician, Instrumentation Technician and related titles. Local availability must be verified.

Canada — NOC 22310

Canada classifies the comparison occupation as **NOC 22310 — Electrical and electronics engineering technologists and technicians**.

Current national Job Bank wages, updated November 19, 2025, are:

- low: **C\$24.04/hour**;
- median: **C\$35.58/hour**;
- high: **C\$55.34/hour**.

Job Bank says technologists usually require a two- or three-year college program; technicians usually require a one- or two-year program.

Certification through provincial associations may be required for some jobs, and title regulation varies by province. Québec and Saskatchewan have specific regulated-title considerations identified by Job Bank. Verify the province before relying on a title or credential.

Canada's labour-market agreements and training gateways can support eligible learners, but funding is not automatic.

Colombia — CUOC 31141

Colombia maps the core comparison to **CUOC 31141 — Técnicos en electrónica**, competence level 3.

OCUPACOL describes work supporting electronic/electrical engineering research, design, development, manufacturing, assembly, integration, programming, operation, testing, inspection, maintenance and repair.

Listed titles include:

- Técnico de electrónica;
- Técnico de electrónica industrial;
- Técnico de ingeniería electrónica;
- Técnico de ingeniería de instrumentación;
- Técnico de mantenimiento electrónico;
- Tecnólogo en electrónica;
- Tecnólogo en mantenimiento electrónico e instrumental industrial.

Related occupation **CUOC 31142** covers automation and instrumentation technicians. The classifications overlap in real workplaces but are not interchangeable in every role.

SENA pathway

SENA Betowa currently lists **Mantenimiento electrónico e instrumental industrial** as a **Tecnólogo** titled, presencial program with a displayed duration of **24 months** on the reviewed program page.

The current program description includes preventive/corrective maintenance, monitoring variables, failure prediction, inspection and maintenance planning for electronic/instrumentation systems.

The reviewed offer lists 11th-grade education and selection/aptitude requirements. Cohort and location availability change, so verify the live Betowa listing before applying.

SENA also lists complementary electronics learning such as **Electrónica básica** in selected cohorts. Complementary study is useful but is not equivalent to a technologist program or regulated electrical authorization.

Latin America and Caribbean training locator

OIT/Cinterfor links vocational-training institutions across Latin America and the Caribbean.

Useful search terms include:

- electrónica;

- electrotecnia;
- instrumentación;
- automatización;
- mecatrónica;
- mantenimiento electrónico;
- controles industriales;
- calibración.

Country institutions, credentials and funding rules differ. Verify locally.

Brazil — SENAI route

SENAI São Paulo currently lists **Técnico em Eletrônica** as a **1,200-hour** presencial technical program.

Its stated objective includes developing electronic circuits, integrating systems and supervising equipment maintenance while considering technical, quality, health, safety and environmental requirements.

Brazilian CBO includes electronics/electrotechnics technician occupations such as **3132-15 Técnico eletrônico** and related codes. Verify current education and professional-practice requirements for the specific state and job.

Peru — SENATI locator

SENATI provides industrial technical training in areas such as:

- Electrónica Industrial;
- Electrotecnia Industrial;
- Mecatrónica Industrial;
- Electricista Industrial;
- automation and control-related fields.

Verify the current program page for duration, campus, admission and fees. Do not rely on an old catalog as proof that a current seat exists.

Cybersecurity for connected equipment

Modern electronics and industrial equipment may contain:

- network interfaces;
- firmware;
- configuration files;
- user accounts;
- remote-access functions;
- removable media;

- diagnostic software;
- equipment inventories;
- logs and production data.

Good practice includes:

- unique passwords and approved password managers;
- multifactor authentication where supported;
- software/firmware updates through approved processes;
- least-privilege accounts;
- approved removable media;
- configuration backups;
- change records;
- phishing awareness;
- verification of unexpected access requests.

Do not connect personal devices, install unauthorized software, bypass security controls or upload proprietary system information to unapproved services.

Responsible use of AI

AI can support low-risk learning and documentation when employer policy allows it.

Potential uses:

- explaining terminology;
- generating practice questions;
- organizing non-sensitive notes;
- drafting a first-pass troubleshooting checklist for human review;
- suggesting test cases for a simulated circuit;
- helping compare two approved documentation revisions;
- translating plain-language study notes for review.

AI must not be used as:

- a voltage or current measurement;
- a calibration standard;
- a protection setting authority;
- engineering approval;
- an energized-work decision maker;
- a substitute for manufacturer documentation;
- a substitute for qualified-person electrical safety procedures.

Verify consequential output against approved schematics, specifications, measurements, calibration records, manufacturer documentation and responsible technical authority.

Accessibility and practical learning

Technical learning should be rigorous without creating unnecessary barriers.

Useful practices include:

- short modules;
- clear diagrams with text explanations;
- high-contrast materials;
- captions/transcripts for video;
- keyboard-accessible software where available;
- checklists for multistep procedures;
- extra time or alternate demonstration methods when appropriate;
- simulation and deenergized lab alternatives when a field environment is unnecessary for the learning objective.

Accommodation does not remove essential safety, qualification or professional requirements.

Building a safe entry-level portfolio

A portfolio can demonstrate thinking without using live or confidential equipment.

Possible items:

1. a schematic-reading exercise with annotated signal flow;
2. a synthetic troubleshooting tree for a low-energy training circuit;
3. a mock test report with expected/actual/result fields;
4. a component-identification sheet;
5. a simulated calibration record showing traceability concepts;
6. a revision-controlled maintenance checklist;
7. a small microcontroller or educational electronics project using manufacturer-safe voltage/power limits;
8. a cybersecurity/change-control checklist for a fictional connected controller;
9. a fault-analysis report using invented data clearly labeled synthetic.

Do not publish confidential schematics, firmware, customer data, security architecture, credentials, equipment serials or proprietary procedures.

Resume strategy

Entry-level resumes should emphasize evidence rather than inflated titles.

Useful evidence may include:

- associate/technical program coursework;
- supervised lab work;
- schematic reading;
- component identification;
- approved test-equipment experience;
- soldering/rework training;
- instrumentation/controls exposure;
- PLC or automation coursework;
- documentation and QA;
- calibration awareness;
- preventive maintenance;
- cybersecurity habits;
- safe work practices;
- projects using synthetic or authorized material.

State proficiency honestly. “Coursework in PLC fundamentals” is better than claiming “PLC engineer” after one class.

Entry-level interview preparation

Be ready to explain:

- how you confirm the correct schematic revision;
- how you approach a fault without replacing parts randomly;
- how you document measurements;
- how you respond when a result conflicts with expectations;
- when you stop and escalate for safety;
- the difference between technician support and engineering approval;
- why energized testing is restricted;
- how you protect configuration files and credentials;
- how you verify AI-assisted technical text;
- how you learn unfamiliar equipment.

Strong answers describe repeatable processes, not bravado.

Six-step action plan

Step 1 — Confirm the local occupation and training route

- Review O*NET 17-3023.00 or your country’s equivalent.
- Save five local job postings.

- Highlight recurring education, equipment, software, safety and credential requirements.
- Identify any licensing or title restrictions.

Milestone: one-page local requirements summary.

Step 2 — Build electrical/electronics fundamentals safely

Practice with simulations or purpose-built educational hardware:

- units and algebra;
- voltage/current/resistance/power concepts;
- schematic symbols;
- analog/digital basics;
- components;
- measurement theory;
- documentation.

Milestone: one completed fundamentals checklist plus a synthetic schematic-reading exercise.

Step 3 — Add test and troubleshooting discipline

Learn:

- instrument purpose and limits;
- calibration status awareness;
- expected-versus-actual thinking;
- fault isolation;
- revision control;
- safe stop/escalation rules.

Milestone: one synthetic troubleshooting report using a deenergized or simulated system.

Step 4 — Choose a specialization

Select one starting direction:

- electronics manufacturing;
- instrumentation/controls;
- field service;
- communications;
- automation/mechatronics;
- calibration/test;
- utilities;
- laboratory/R&D support.

Research the tools, credentials and safety requirements for that direction.

Milestone: one specialization map listing skills, training and three target employers.

Step 5 — Build a safe portfolio and targeted resume

- Use only synthetic, educational or authorized material.
- Include three to five evidence-based samples.
- Show documentation and QA habits.
- Describe tools by actual proficiency.
- Do not claim engineering, electrician, energized-work or certification authority you do not have.

Milestone: portfolio folder plus a one-page targeted resume.

Step 6 — Apply, interview and improve

- Apply to roles whose entry requirements you substantially meet.
- Track recurring skill gaps.
- Practice process-based interview answers.
- Ask employers about supervision, training, energized-work policy, calibration, documentation and cybersecurity expectations.
- Update your learning plan every week.

Milestone: a four-week application/learning log with weekly adjustments.

A four-week starter schedule

Week 1

- review the occupation profile;
- learn units and schematic symbols;
- complete basic simulation exercises;
- research three local training programs.

Week 2

- practice reading wiring/schematic diagrams;
- learn component functions;
- study instrument purpose/rating concepts;
- create one mock test report.

Week 3

- choose a specialization;
- build one safe portfolio exercise;
- practice a structured troubleshooting scenario;
- draft your resume.

Week 4

- finish three portfolio samples;
- review ten local job postings;
- tailor your resume to recurring requirements;
- practice interview responses;
- submit appropriate applications;
- record the next skill gaps.

Questions to ask a school or training provider

Before paying or enrolling, ask:

- Is the program electrical engineering technology, electronics engineering technology, electronics technology, instrumentation or another field?
- What credential is awarded?
- Is the program recognized by employers in my region?
- What lab equipment is available?
- Are labs designed around safe training voltages and controlled procedures?
- How are energized-work topics handled?
- Are calibration, documentation and QA taught?
- Are automation/PLC/instrumentation topics included?
- Is cybersecurity covered for connected equipment?
- Is work-based learning available?
- What is the total cost, including tools and fees?
- Is WIOA, provincial support, SENA or other funding available and am I eligible?
- What completion and job-placement data can the provider document?

Questions to ask an employer

At interview or onboarding, clarify:

- What equipment will I support?
- Who provides engineering direction?
- Which tasks require a licensed electrician or professional engineer?
- Does the role involve energized testing?
- What qualifies a worker for electrical testing at this site?
- What lockout/tagout program is used?
- Which test instruments require calibration?
- How are schematics and revisions controlled?
- What training is provided before independent work?
- What cybersecurity/change-control rules apply?

- Is AI use allowed, restricted or prohibited?
- What advancement paths exist for technicians?

Source links for reader verification

- O*NET occupation: <https://www.onetonline.org/link/details/17-3023.00>
- O*NET wages: <https://www.onetonline.org/link/localwages/17-3023.00>
- BLS occupation: <https://www.bls.gov/ooh/architecture-and-engineering/electrical-and-electronics-engineering-technicians.htm>
- CareerOneStop WIOA finder: <https://www.careeronestop.org/LocalHelp/EmploymentAndTraining/find-WIOA-training-programs.aspx>
- Canada Job Bank summary: <https://www.jobbank.gc.ca/marketreport/summary-occupation/18017/ca>
- Canada Job Bank wages: <https://www.jobbank.gc.ca/marketreport/wages-occupation/18017/ca>
- Canada Job Bank requirements: <https://www.jobbank.gc.ca/marketreport/requirements/18017/ca>
- Canada training gateway: <https://www.canada.ca/en/services/jobs/training.html>
- Canada labour-market agreements: <https://www.canada.ca/en/employment-social-development/programs/training-agreements.html>
- Colombia OCUPACOL: <https://ocupacol.mintrabajo.gov.co/Profile/OccupationalProfile/31141>
- SENA maintenance/instrumentation: <https://betowa.sena.edu.co/oferta/mantenimiento-electronico-e-instrumental-industrial?modality=P&programId=69>
- SENA basic electronics: <https://betowa.sena.edu.co/oferta/electronica-basica?modality=P&programId=13847>
- OIT/Cinterfor: <https://www.oitcinterfor.org/>
- SENAI Técnico em Eletrônica: <https://www.sp.senai.br/curso/tecnico-em-eletronica/107711>
- SENATI: <https://www.senati.edu.pe/>
- OSHA 1910.333: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.333>
- OSHA 1910.334: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.334>
- CISA Secure Our World: <https://www.cisa.gov/secure-our-world>
- NIST AI RMF: <https://www.nist.gov/itl/ai-risk-management-framework>

- NIST Generative AI Profile:
<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- U.S. Indeed Electrical Technician context:
<https://www.indeed.com/career/electrical-technician/salaries>
- U.S. Indeed Electronics Technician context:
<https://www.indeed.com/career/electronics-technician/salaries>
- Colombia Indeed Técnico electrónico context:
<https://co.indeed.com/career/t%C3%A9cnico-electronico/salaries>

Important limits

This guide provides educational and career-planning information. It does not guarantee:

- employment;
- wages or salary;
- admission;
- funding;
- apprenticeship placement;
- certification;
- licensing;
- professional engineering authority;
- electrician authority;
- qualified-person status;
- permission to perform energized testing;
- calibration authority;
- project or safety approval.

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